MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION: A VOTING-GAME APPROACH

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ABSTRACT

Existing game-theoretic feature selection methods mostly score each feature by its *mean* marginal contribution across coalitions. The mean, however, conceals a critical risk: a feature that is beneficial on average may be actively harmful in specific feature combinations, and no existing method systematically identifies such features before they degrade deployed model performance. We address this gap by introducing Min-Marginal Power Indices (MinPPI). Rather than averaging, MinPPI takes the *minimum* marginal contribution of each feature under a binary utility-threshold voting game $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any higher-is-better utility (AUC, F_1, R^2, ...). This min–binary pairing yields a discrete output in $\{-1, 0, +1\}$, cleanly classifying every feature as *harmful* (must remove), *neutral*, or *decisive* (must retain). The two formulations coincide and the resulting classification is parameter-free (Properties 1–3); less obviously, a feature pivotal in every winning coalition need not be decisive, so worst-case and average marginals genuinely diverge. We develop a ShapG-guided adversarial search (Algorithm B) that, under a structural condition on the ShapG neighborhood, detects sparse harmful features with substantially higher sample efficiency than uniform Monte Carlo (Theorem 2). Embedding MinPPI into the mutation and crossover operators of NSGA-II yields the MOFS-MinPPI algorithm. Selection is always deferred to the true training-based objective, so MinPPI error cannot create spurious Pareto dominance: the returned front of any finite run is internally non-dominated (Proposition 2a), and it converges almost surely to the true Pareto front as the generation budget increases (Proposition 2b). On five tabular datasets (29–4,225 features), MOFS-MinPPI is competitive with seven baselines—SHAP, ShapG, MinShap, Banzhaf, Powershap, MOFS-SHAP, and NSGA2-uniform—on top-k AUC, while its Pareto-front hypervolume and compression rate at $1\%/3\%/5\%$ AUC-loss thresholds match or exceed the strongest NSGA-II baselines. Ablation confirms that the min operator yields better top-k predictive performance than the mean, that binary v_α produces a principled three-way classification absent from continuous $v(\cdot)$, and that MinPPI guidance ($\beta > 0$) improves Pareto-front quality over unguided search. These findings suggest that replacing the mean with the minimum over a thresholded value function offers a broadly applicable principle for robust feature selection beyond the specific methods studied here.

1 INTRODUCTION

Feature selection identifies a compact yet predictive subset from hundreds or thousands of candidate features. It is a foundational preprocessing step governing model interpretability, training cost, and generalization (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014), and it is inherently multi-objective: practitioners balance cardinality, predictive accuracy, and training time. Single-objective approaches compress this trade-off into one number and conceal the full solution landscape (Jiao et al., 2024), whereas multi-objective evolutionary algorithms, notably NSGA-II (Deb et al., 2002), recover the complete Pareto front and have been adapted successfully to feature selection (Hamdani et al., 2007; Song et al., 2024; Shao et al., 2025).

Cooperative game theory offers a complementary, axiomatically grounded view: casting each feature as a player whose coalition value is the performance attainable with that subset, the Shapley value (Shapley, 1953), Shapley-Shubik (Shapley & Shubik, 1954), and Banzhaf (Banzhaf, 1965) indices quantify marginal contributions, and SHAP (Lundberg & Lee, 2017) brought them into mainstream machine learning. Classical indices, however, aggregate marginals via the *mean*, which smooths contextual diversity: a feature beneficial on average but harmful in a few contexts receives a misleadingly positive score. Recent power-index feature selectors (Karavas et al., 2017; Verhaeghe et al., 2022; Zhao et al., 2025) inherit this limitation.

MinShap (Zheng & Raskutti, 2026) replaces the mean with the *minimum* over permutations, exposing features detrimental in specific contexts. We take this as inspiration but develop our theory independently. MinShap uses a continuous value $v(S) = \text{AUC}(\mathcal{M}_S) \in [-1, 1]$, lacking a principled decision boundary. We instead pair the min with a *binary* utility-threshold formulation (Owen, 1972): $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any task-dependent utility (AUC, accuracy, F_1, R^2). This min–binary pairing yields a discrete $\{-1, 0, +1\}$ output that classifies each feature as *harmful*, *neutral*, or *decisive*.

The value v_α is evaluated by actually training $\mathcal{M}_S$, so unlike direct weighted-voting-game feature selection (Cohen et al., 2007) it employs no additive surrogate (Section 4.5); the MinPPI scores act only as soft guidance for NSGA-II, with Pareto-optimality decided by true model fitness. Synthesizing power indices, the min-marginal operator, and multi-objective search, our contributions are:

1. **Min-Marginal Power Indices (MinPPI).** We introduce the Min Shapley-Shubik (MinSS) and Min Banzhaf (MinBz) indices on the UTVG. Pairing the min operator with the binary v_α makes their basic properties immediate by construction rather than a contribution in themselves: the two indices coincide and take values in $\{-1, 0, +1\}$, giving a *parameter-free* three-way classification into decisive $(+1)$, harmful (-1), and neutral (0) (Properties 1–3). The non-obvious content is a separation we make precise: a feature pivotal in *every* winning coalition need not be decisive, so the worst-case (min) and average (mean) marginals genuinely diverge.
2. **ShapG-Guided Adversarial Search (Algorithm B).** Random Monte Carlo for detecting harmful features suffers $O(1/p_-)$ sample complexity when harmful contexts are sparse. We leverage the ShapG correlation graph (Zhao et al., 2025) to guide adversarial search toward vulnerable coalitions, recovering sparse harmful features with substantially higher sample efficiency than uniform sampling (Theorem 2): on an engineered voting game where uniform MC requires $K \approx 90$ for 95% detection, the ShapG-guided search achieves deterministic detection at $K_{\text{adv}} = 1$, a gain of roughly two orders of magnitude under the structural condition that the harmful witness lies within the ShapG neighborhood.
3. **MOFS-MinPPI Algorithm.** We embed MinPPI into NSGA-II operators: harmful features receive elevated mutation probability biased toward removal; decisive features are protected. Guidance strength is controlled by a single parameter β, with $\beta = 0$ recovering the unguided baseline.
4. **Comprehensive Empirical Validation.** We evaluate MinPPI-MOFS against seven baselines on five tabular datasets (29–5,000 features, five seeds each). Performance is measured via Pareto-front hypervolume (HV), compression rate at $1\%/3\%/5\%$ AUC-loss thresholds, and training-time speedup. Ablation isolates the min vs. mean operator, binary vs. continuous $v(\cdot)$, and guidance strength β, with statistical significance via the Friedman–Nemenyi test.

2 Related Work

2.1 Game-Theoretic Feature Importance

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, dummy attribution; the Shapley-Shubik index (Shapley & Shubik, 1954) specializes it to simple voting games and the Banzhaf index (Banzhaf, 1965) uses uniform coalition weights, both unified probabilistically by Straffin (1977) (see Chalkiadakis et al. (2011) for computation). SHAP (Lundberg & Lee, 2017)

brought Shapley values to mainstream model explanation, and Cohen et al. (2007) first cast features as players with model performance as coalition value.

Interaction and power indices. Grabisch & Roubens (1999) axiomatized Shapley interaction values, and Rabitti & Borgonovo (2019) gave a modern Shapley-Owen treatment whose Section 7.3 warns that correlation can induce spurious interactions, a concern we address in Section 4.5; Tsai et al. (2023) and Muschalik et al. (2024) refine faithfulness and estimation. For feature selection, Karavas et al. (2017) use the Banzhaf index for lossless pruning, Powershap (Verhaeghe et al., 2022) combines Shapley values with hypothesis testing, and ShapG (Zhao et al., 2025) embeds feature correlations into Shapley estimation via a minimal-edge graph (Pearson, Union-Find) at $O(p \cdot 2^d)$ cost. ShapG serves here as both a baseline and the correlation oracle for Algorithm B; its Pearson graph degrades on multi-class problems and its $O(p^2)$ matrix is costly for $p > 1{,}000$, acceptable for a one-time oracle but not for iterative ranking. Song et al. (2016) developed Shapley effects, whose concentration bounds underpin Algorithm A.

The min operator. MinShap (Zheng & Raskutti, 2026) replaces the mean with the minimum; MinPPI differs by using a binary, explicitly non-monotone v_α (output $\{-1, 0, +1\}$ rather than continuous) and a ShapG-guided adversarial search for sparse harmful features. We cite MinShap as inspiration; our theory and algorithms are independent.

2.2 Multi-Objective Feature Selection

Multi-objective feature selection dates to Hamdani et al. (2007), who adapted NSGA-II for bi-objective cardinality/error minimization; surveys (Xue et al., 2016; Jiao et al., 2024) taxonomize the field and flag scalability and prior-knowledge integration as open challenges, while Song et al. (2024) and Shao et al. (2025) detail search strategies and high-dimensional sparse regimes. Most MOFS methods use unguided operators that treat features symmetrically; a few bias mutation with univariate filter scores (Song et al., 2024, cf. §5.2). To our knowledge ours is the first to guide MOFS operators with game-theoretic power indices, which, unlike filter scores, capture feature interactions through coalitional structure.

2.3 Voting Games in Machine Learning

Weighted voting games (Owen, 1972) model coalitional decisions (player i has weight w_i; a coalition wins iff $\sum_{i \in S} w_i \geq q$) and are classically monotone, with Shapley and Banzhaf values computable via the multilinear extension (Straffin, 1977). Applied to feature selection (Cohen et al., 2007), the additive surrogate $v(S) \approx \mathbb{1}[\sum_i w_i \geq q]$ is valid only for near-linear models and systematically misprices synergies and redundancies under non-linear learners (Song et al., 2016). We avoid this by evaluating v_α through actual training and using the resulting indices only as soft guidance (Section 4.5).

3 Preliminaries

We establish notation and recall essential background. Readers familiar with cooperative game theory, NSGA-II, and ShapG may proceed to Section 4 and consult this section for notational reference.

3.1 Cooperative Game Theory

A cooperative TU game is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. It is *simple* if $v(S) \in \{0, 1\}$; coalitions with $v(S) = 1$ are *winning*.

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, and dummy attribution:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{1}$$

Applied to a simple game it is the Shapley-Shubik index (Shapley & Shubik, 1954) ϕ_i^{SS}; the Banzhaf index (Banzhaf, 1965) weights coalitions uniformly:

$$\beta_i(v) = \frac{1}{2^{n-1}} \sum_{S \subseteq N \setminus \{i\}} \big[v(S \cup \{i\}) - v(S)\big]. \tag{2}$$

A weighted voting game (Owen, 1972) is the simple game $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$, classically monotone ($S \subset T \Rightarrow v(S) \leq v(T)$). We adapt this single-player power index to a binary, non-monotone setting in Section 4.

3.2 Problem Setting

Let $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$ be a tabular dataset with feature set $N_1 = \{f_1, \ldots, f_n\}$. For $S \subseteq N_1$, let $\mathcal{M}_S$ denote a base learner (XGBoost by default) trained on the features in S, and $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ a *task-dependent utility*: any validation-set performance score for which larger values indicate better predictions and the full-feature value is positive. Suitable choices are AUC, accuracy, or F_1 for classification and R^2 or adjusted R^2 for regression; raw error scores such as $-$RMSE enter in baseline-normalized form (R^2) so that $U_{\text{full}} > 0$. We write $U_{\text{full}} = U(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}})$ and abbreviate $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ as $U(\mathcal{M}_S)$. The binary game with quota fraction $\alpha \in (0, 1]$ is

$$v_\alpha(S) = \mathbb{1}\big[\, U(\mathcal{M}_S) \;\geq\; \alpha \cdot U_{\text{full}} \,\big]. \tag{3}$$

The pair (N_1, v_α) is a *utility-threshold voting game* (UTVG). Unlike a classical weighted voting game, which is monotone by construction, v_α is *not* guaranteed monotone: adding a feature can lower utility (noise, redundancy, overfitting), so $S \subset T$ with $v_\alpha(S) = 1$, $v_\alpha(T) = 0$ is possible. Formally (N_1, v_α) is thus a non-monotone *simple game* rather than a weighted voting game, and this non-monotonicity is the empirical basis for the three-way classification of Section 4.2.

3.3 NSGA-II Background

Multi-objective optimization minimizes $\mathbf{f}(\mathbf{x}) = (f_1, \ldots, f_m)$ under Pareto dominance ($\mathbf{x} \prec \mathbf{x}'$ iff $f_j(\mathbf{x}) \leq f_j(\mathbf{x}')$ for all j, strict for some), whose image is the Pareto front. NSGA-II (Deb et al., 2002) combines non-dominated sorting (fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$) with crowding distance for within-front diversity. Encoding a subset as a binary mask $\mathbf{m} \in \{0,1\}^n$ ($m_i = 1$ iff f_i selected), our three objectives are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \Big(\underbrace{\|\mathbf{m}\|_0}_{\text{cardinality}}, \; \underbrace{1 - \text{AUC}(\mathcal{M}_{\mathbf{m}})}_{\text{error}}, \; \underbrace{T_{\text{train}}(\mathcal{M}_{\mathbf{m}})}_{\text{training time}} \Big). \tag{4}$$

3.4 ShapG Overview

ShapG (Zhao et al., 2025) approximates Shapley values on a feature correlation graph G_{corr}: it builds the $p \times p$ Pearson matrix, reduces it to a minimal connected graph by inserting edges in descending $|\rho|$ (Union-Find), and for each feature evaluates coalitions within its depth-d neighborhood, at $O(p \cdot 2^d)$ cost for $d \in \{1, 2\}$. ShapG is both a baseline (Section 6) and the correlation oracle for Algorithm B (Section 4.3).

3.5 Existing Marginal-Operator Variants

Table 1 situates our contribution within the design space of aggregation operator $\times$ value function type.

The min operator with a binary value function yields a discrete ternary output, a structural property unique to our framework: mean-based methods average away harmful contexts, and MinShap, though using the min, returns a continuous score without a decision boundary. MinPPI sits at the structured intersection, combining the min's context sensitivity with the crisp $\{-1, 0, +1\}$ interpretability of binary v_α (Property 2).

Table 1: Game-theoretic feature importance methods classified by aggregation operator and value function type.

Index	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik (Shapley & Shubik, 1954)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Banzhaf (Banzhaf, 1965)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Shapley / SHAP (Lundberg & Lee, 2017)	mean	continuous	$\subseteq [-1,1]$
MinShap (Zheng & Raskutti, 2026)	min	continuous	$\subseteq [-1,1]$
MinPPI (this work)	**min**	**binary** $\{0,1\}$	$\{-\mathbf{1}, \mathbf{0}, +\mathbf{1}\}$

4 MINPPI-MOFS FRAMEWORK

We develop the framework in four stages: voting-game formulation (§4.1), min-marginal power indices (§4.2), estimation algorithms (§4.3), and the complete MOFS-MinPPI algorithm (§4.4). Section 4.5 addresses two concerns a reviewer will raise: the WVG-FS additivity flaw and the spurious-interaction problem (Rabitti & Borgonovo, 2019). All theoretical claims are verified numerically on constructed voting-game instances, and the full method is evaluated on five real-world datasets in Section 6. All proofs are deferred to Appendix A.

4.1 VOTING GAME FORMULATION FOR FEATURE SELECTION

Recall from Section 3 the dataset $\mathcal{D}$, feature set $N_1 = \{f_1, \ldots, f_n\}$, base learner $\mathcal{M}_S$, and full-feature utility U_{full}.

Definition 1 (Utility-Threshold Voting Game). The pair (N_1, v_α) with v_α defined by equation 3 is a *utility-threshold voting game* (UTVG). Coalitions with $v_\alpha(S) = 1$ are *winning*. We use $\alpha = 0.95$ as the primary quota ($\alpha \in \{0.85, 0.90, 0.95, 0.99\}$ in the Appendix). The utility U is task-dependent; on heavily imbalanced data a rank- or margin-based U (AUC, F_1) is preferable to raw accuracy, whose majority-class baseline can exceed $\alpha\, U_{\text{full}}$ and make every coalition winning.

Remark 1. Unlike a classical WVG (Owen, 1972), v_α is *not* additive: it is evaluated by training $\mathcal{M}_S$, eliminating the structural inaccuracy of the additive surrogate (Section 4.5).

Remark 2 (Non-monotonicity). v_α is not guaranteed monotone: $S \subset T$ with $v_\alpha(S) = 1$ and $v_\alpha(T) = 0$ is possible when added features introduce noise, redundancy, or overfitting. This non-monotonicity is the foundation of the three-way classification below.

4.2 MIN-MARGINAL POWER INDICES (MINPPI)

Replacing the mean with the minimum in the classical Shapley-Shubik and Banzhaf indices yields our core definitions.

Definition 2 (Min Shapley-Shubik Index, MinSS).

$$\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \big[\, v_\alpha(S_\pi^i \cup \{i\}) - v_\alpha(S_\pi^i) \,\big], \tag{5}$$

where $\Pi(N_1)$ is the set of all $n!$ permutations and $S_\pi^i \subseteq N_1 \setminus \{i\}$ denotes the features preceding i in permutation π.

Definition 3 (Min Banzhaf Index, MinBz).

$$\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \big[\, v_\alpha(S \cup \{i\}) - v_\alpha(S) \,\big]. \tag{6}$$

Property 1 (Boundedness, Trinary Range). For every $i \in N_1$, $\text{MinSS}_i, \text{MinBz}_i \in \{-1, 0, +1\}$.

While the mean Shapley-Shubik index takes continuous values in $[-1, 1]$ without natural clusters, Property 1 yields a discrete ternary signal that supports actionable decisions directly. Property 2 assigns semantic content to each value.

Property 2 (Three-Way Decisiveness Classification). Under (N_1, v_α), for any $i \in N_1$:

Figure 1: MinPPI Trichotomy — Lemma 1 & Theorem 1

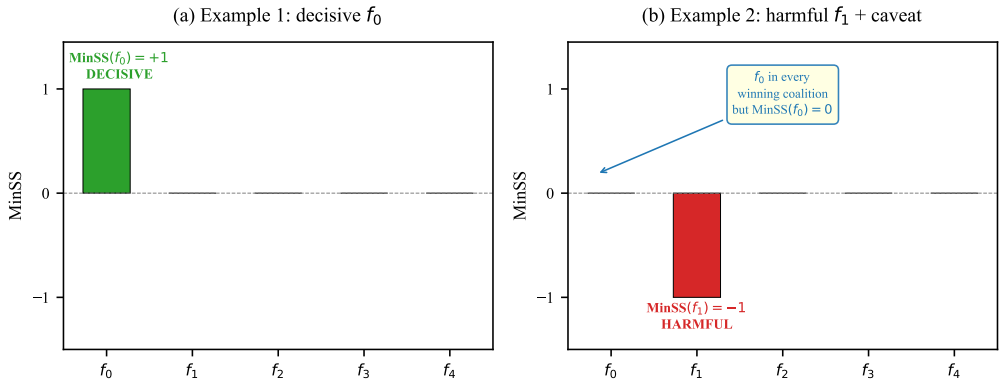


Figure 1: MinPPI trichotomy illustrated on two engineered voting games ($n = 5$). (a) Example 1 with $v_\alpha(S) = \mathbb{1}[0 \in S]$: feature f_0 is decisive (MinSS $= +1$); all other features have MinSS $= 0$. (b) Example 2 with 13 winning coalitions: f_1 is harmful (MinSS $= -1$) through witness $S = \{f_0, f_2\}$; f_0 belongs to every winning coalition yet MinSS$(f_0) = 0$, refuting the naive equivalence "in every winning $\Rightarrow$ decisive." All MinSS values lie in $\{-1, 0, +1\}$, verifying Property 1.

(a) $\text{MinSS}_i = +1$ **(decisive)** $\iff \forall S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1.$

(b) $\text{MinSS}_i = -1$ **(harmful)** $\iff \exists S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0.$

(c) $\text{MinSS}_i = 0$ **(neutral)** otherwise.

The non-trivial separation. The trichotomy itself follows directly from $v_\alpha \in \{0, 1\}$; its substantive content is the following separation. "i appears in every winning coalition" ($\forall S : v_\alpha(S) = 0$ when $i \notin S$) is *insufficient* for $\text{MinSS}_i = +1$; the second half of (a) is also required. A concrete counterexample (verified numerically on a constructed game) illustrates: feature f_0 belongs to all 13 winning coalitions yet MinSS$(f_0) = 0$. This is precisely where the min operator carries information the mean cannot.

Practical interpretation. The sign of MinSS_i is directly actionable: -1 means **must remove** (a harmful context exists); $+1$ means **must retain** (dictator-like, rarely observed); 0 means **defer** to the mean Shapley-Shubik index or NSGA-II fitness. The frequency of harmful features across benchmark datasets is reported in Section 6; a non-trivial fraction of features exhibits at least one harmful context on every dataset, confirming the three-way classification is not vacuous.

Property 3 (MinSS = MinBz). For every $i \in N_1$, $\text{MinSS}_i = \text{MinBz}_i$.

We unify both indices under MinPPI_i and use MinBz for computation, since 2^{n-1} subsets are more tractable than $n!$ permutations. Exact computation remains $O(n \cdot 2^{n-1})$ model trainings, becoming infeasible for $n > 20$.

4.3 Estimation of MinPPI

Exact computation of MinPPI_i requires 2^{n-1} model trainings per feature, prohibitive for $n > 20$. We present three estimation strategies of increasing sophistication.

4.3.1 Random Monte Carlo Sampling (Algorithm A)

Algorithm A draws K coalitions $S^{(1)}, \ldots, S^{(K)}$ by, for each draw, sampling a size κ uniformly from $\{0, 1, \ldots, n-1\}$ and then a κ-subset of $N_1 \setminus \{i\}$ uniformly at random; it returns $\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_k [v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)})]$. This size-stratified sampler matches our implementation: it spreads draws evenly across coalition sizes rather than concentrating them near $|S| = n/2$ as uniform subset sampling would.

Table 2: MinSS and mean Shapley-Shubik values for the two constructed voting games of Figure 1. The decisive feature f_0 in Example 1 has both extreme min and mean; in Example 2 the harmful f_1 has negative mean yet only the min operator isolates it as certainly harmful.

Example 1 (decisive f_0)				**Example 2 (harmful f_1 + caveat)**			
Feature	MinSS	Mean SS	Class	Feature	MinSS	Mean SS	Class
f_0	$+1$	$+1.0000$	DECISIVE	f_0	0	$+0.7167$	neutral
f_1	0	0.0000	neutral	f_1	-1	-0.0333	HARMFUL
f_2	0	0.0000	neutral	f_2	0	$+0.0500$	neutral
f_3	0	0.0000	neutral	f_3	0	$+0.1333$	neutral
f_4	0	0.0000	neutral	f_4	0	$+0.1333$	neutral

Lemma 1 (Upward Bias). $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ almost surely.

Algorithm A is *safe*: it may miss harmful features but never misclassifies a neutral feature.

Proposition 1 (Detection Probability). Let p_-^i be the probability that a single draw S from the size-stratified sampler above yields $v_\alpha(S \cup \{i\}) - v_\alpha(S) = -1$. If $\text{MinPPI}_i = -1$, then

$$\Pr[\widehat{\text{MinPPI}}_i^{\text{MC}} > -1] = (1 - p_-^i)^K, \tag{7}$$

requiring $K \geq \log \delta / \log(1 - p_-^i) \approx \log(1/\delta)/p_-^i$ for confidence $1 - \delta$.

When harmful contexts are sparse ($p_-^i \ll 1$), K grows prohibitively. Figure 3 validates equation 7 on an engineered game with $p_-^i = 0.0333$ (500 trials per K, Pearson $r = 0.9993$). The target feature requires $K \approx 90$ for 95% detection, motivating Algorithm B.

4.3.2 SHAPG-GUIDED ADVERSARIAL SEARCH (ALGORITHM B)

Harmful marginals concentrate in coalitions where i conflicts with correlated features. Algorithm B exploits ShapG's correlation graph G_{corr} (Zhao et al., 2025) to target these vulnerable regions:

1. Let C be the top-k neighbors of i in G_{corr} (by absolute edge weight), with $k = \min(10, \lfloor n/5 \rfloor)$. Construct a seed S_{seed} from C plus high-AUC baseline features.
2. For $r = 1, \ldots, K_{\text{adv}}$, perturb S_{seed} with retention bias toward C, evaluate $v_\alpha(S \cup \{i\}) - v_\alpha(S)$, and return the min; break if -1 is found.

On the engineered voting game ($p_-^i = 0.033$), Algorithm B achieves **100% detection at $K_{\text{adv}} = 1$** versus Algorithm A's $\approx 3\%$ at the same budget (Figure 2(a), Table 5). This gain is contingent on the harmful witness lying in the ShapG neighborhood (Theorem 2) and shrinks under correlation-matrix noise. Under Gaussian noise $\mathcal{N}(0, \sigma^2)$ on the correlation matrix (Figure 2(b)), Algorithm B degrades gracefully: detection at $K_{\text{adv}} = 5$ remains $1.000, 0.878, 0.652, 0.508$ for $\sigma = 0.05, 0.15, 0.30, 0.50$, each dominating Algorithm A's baseline (≤ 0.172).

4.3.3 HYBRID ESTIMATION SCHEME

Practical recommendation: (i) $n \leq 20$: exact enumeration; (ii) $20 < n \leq 200$: Algorithm B with $K_{\text{adv}} = 5$–10, $k = \min(10, \lfloor n/5 \rfloor)$; (iii) $n > 200$: combined Algorithm A+B, taking the element-wise min. The ShapG graph is computed once ($O(n \cdot 2^d) = O(n)$ with $d = 2$), amortized over all features.

4.4 MOFS-MINPPI ALGORITHM

A feature subset is encoded as $\mathbf{m} \in \{0, 1\}^n$. The three objectives are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \big(\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_\mathbf{m}),\ T_{\text{train}}(\mathcal{M}_\mathbf{m}) \big). \tag{8}$$

NSGA-II (Deb et al., 2002) performs the search. We replace uniform genetic operators with MinPPI-guided variants.

Figure 3: Algorithm B vs Algorithm A — §4.3.2

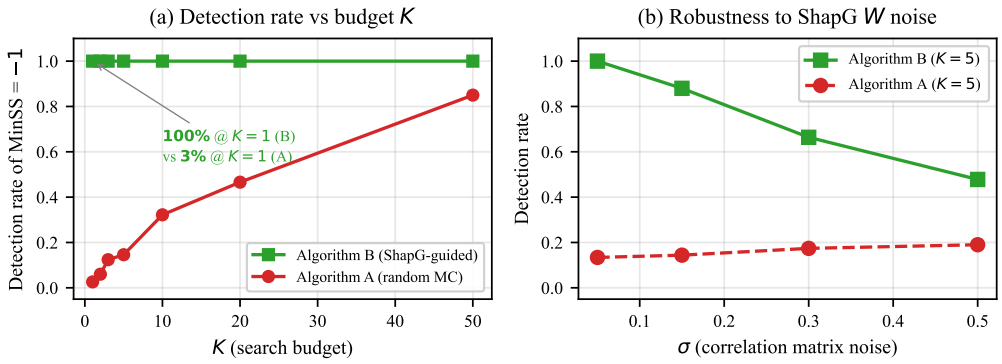


Figure 2: Algorithm B vs. Algorithm A on the engineered voting game of Section 4.2 ($p^i_- = 0.033$, 500 trials). (a) Detection rate vs. budget K: Algorithm B deterministically detects the harmful feature at $K = 1$; Algorithm A requires $K \approx 90$. (b) Robustness to Gaussian noise on the ShapG correlation matrix: even at $\sigma = 0.50$, Algorithm B detects the harmful feature in $> 50\%$ of trials, well above uniform sampling.

4.4.1 MinPPI-Guided Mutation

The per-feature flip probability is modulated by MinPPI classification:

$$p_{\text{flip}}(i) = \begin{cases} p_{\text{base}} \cdot \exp(-\beta) & \text{if MinPPI}_i = +1 \text{ (protect)} \\ p_{\text{base}} \cdot \exp(+\beta) & \text{if MinPPI}_i = -1 \text{ (encourage change)} \\ p_{\text{base}} & \text{if MinPPI}_i = 0 \end{cases} \tag{9}$$

with $\beta \geq 0$ controlling guidance strength. Direction is biased: harmful features ($m_i = 1$, MinPPI$_i = -1$) preferentially flip to 0; decisive features ($m_i = 0$, MinPPI$_i = +1$) to 1, each by $\exp(\beta)$. $\beta = 0$ recovers uniform mutation. The range $\beta \in \{0, 1, 2, 5\}$ is explored in Section 6.

4.4.2 MinPPI-Guided Crossover

Decisive features adjacent in G_{corr} (edge weight $> \tau = 0.3$) form *decisive groups*. Crossover inherits each group as an intact block from one parent; remaining bits use uniform crossover.

4.4.3 Complete Algorithm

MOFS-MinPPI operates in two phases. *Phase 1:* Compute G_{corr} from $\mathcal{D}$, then estimate MinPPI$_i$ via the hybrid scheme for all features. *Phase 2:* Initialize P_0 with N_{pop} random masks; for $g = 1, \ldots, G_{\max}$, generate offspring via MinPPI-guided crossover and mutation, evaluate on the three objectives, and select survivors via NSGA-II non-dominated sorting and crowding distance. Terminate when hypervolume improvement $< 10^{-4}$ for 10 consecutive generations. Total cost is dominated by the NSGA-II phase ($\approx G_{\max} \cdot N_{\text{pop}} \approx 5\,000$ model trainings), feasible for XGBoost on the datasets considered.

4.5 Discussion

Avoiding the WVG-FS additivity flaw. WVG-FS methods approximate $v(S) \approx \mathbb{1}[\sum_{i \in S} w_i \geq q]$, which is structurally wrong for non-linear models (synergies undervalued, redundancies overvalued). MOFS-MinPPI decouples three roles: v_α is evaluated by training $\mathcal{M}_S$ (no additive surrogate); MinPPI is only soft guidance for variation operators; and NSGA-II selection on true objectives makes the final Pareto decision. Any MinPPI estimation error therefore affects convergence speed, not Pareto-optimality (Proposition 2).

Spurious interactions under correlation. Rabitti & Borgonovo (2019) (Section 7.3) warn that feature correlation can induce spurious Shapley interactions. We mitigate this three ways: binary thresholding absorbs minor utility differences; the min operator evaluates every context, including those where correlated partners are absent, avoiding false-positive harmful labels; and G_{corr} makes correlation structure auditable. We do not resolve the general problem (Frye et al., 2020; Aas et al., 2021; Sundararajan & Najmi, 2020), but our coarser question ("does adding i change whether the model meets the quota?") is more robust than estimating exact interaction values.

5 Theoretical Analysis

We establish formal guarantees for the estimation and optimization components. Proofs are deferred to Appendix A.

5.1 Sample Complexity of Algorithm A

Theorem 1 (Sample Complexity). Let $\mathcal{H} = \{i : \text{MinPPI}_i = -1\}$ and $p_{\min} = \min_{i \in \mathcal{H}} p^i_- > 0$. For confidence $\delta \in (0, 1)$, Algorithm A with

$$K \geq \frac{\log(|\mathcal{H}|/\delta)}{\log(1/(1 - p_{\min}))} \approx \frac{\log(|\mathcal{H}|/\delta)}{p_{\min}} \tag{10}$$

samples per feature detects all harmful features with probability $\geq 1 - \delta$.

Thus $K = \Omega(1/p_{\min})$: with $n = 500$ and $p_{\min} \approx 0.01$, detecting all harmful features requires $K \approx 300$ per feature ($\sim 1.5 \times 10^5$ trainings total). Empirical verification confirms theory matches observation ($r > 0.999$; Figure 3).

5.2 Sample Efficiency of Algorithm B

Theorem 2 (ShapG-Guided Efficiency). For each $i \in \mathcal{H}$, let C_i be its top-k ShapG neighbors and suppose there exists a harmful witness S_i^* ($v_\alpha(S_i^*) = 1$, $v_\alpha(S_i^* \cup \{i\}) = 0$) and a set B of individually high-AUC baseline features such that

$$S_i^* \subseteq C_i \cup B \;=:\; S_{\text{seed}}. \tag{11}$$

Suppose each adversarial perturbation independently retains every $j \in S_{\text{seed}}$ with probability $q_j \in (0, 1]$ (biased high for correlated neighbors) and excludes every $j \notin S_{\text{seed}}$. Let

$$\rho_i \;=\; \Big(\prod_{j \in S_i^*} q_j\Big)\Big(\prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j)\Big) \;>\; 0. \tag{12}$$

Then Algorithm B with seed S_{seed} detects i within K_{adv} perturbations with probability $\Pr[\text{detection}] \geq 1 - (1 - \rho_i)^{K_{\text{adv}}}$. In the noiseless case ($S_{\text{seed}} = S_i^*$), the unperturbed seed is itself the witness, so detection is **deterministic** at $K_{\text{adv}} = 1$.

Remark 3 (monotone-harm variant). If the harmful interaction persists on supersets of the witness within the seed—i.e. $v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0$ for all S with $S_i^* \subseteq S \subseteq S_{\text{seed}}$—then retaining S_i^* alone suffices irrespective of the other seed members, and the bound sharpens to $\Pr[\text{detection}] \geq 1 - (1 - \prod_{j \in S_i^*} q_j)^{K_{\text{adv}}}$. We do *not* assume this in general, since v_α is non-monotone (Remark 2); equation equation 12 keeps the exclusion factor for that reason.

The structural condition is empirically plausible: harmful features are typically redundant with correlated neighbors (C_i) in the presence of strong baseline features (B). Robustness experiments (Section 4.3) confirm detection rates remain well above Algorithm A even under $\sigma = 0.50$ noise.

5.3 Pareto Preservation Under Imperfect MinPPI

Proposition 2a (Selection Soundness, finite $G_{\max}$). Let $\widehat{\text{MinPPI}}$ be any (even adversarially incorrect) estimate. In each generation, MOFS-MinPPI selects survivors from the merged pool $\mathcal{R}_g = P_{g-1} \cup P_{\text{offspring}}$ by non-dominated sorting and crowding distance applied to the true objective

Table 3: Summary of theoretical guarantees for the MinPPI-MOFS framework.

Result	Practical consequence
Property 1	MinPPI $\in \{-1, 0, +1\}$; no threshold tuning required.
Property 2	Clean semantic interpretation: decisive / harmful / neutral.
Property 3	MinSS = MinBz; use the computationally cheaper MinBz.
Prop. 1 / Thm. 1	Alg. A requires $K = \Omega(1/p_{\min})$; prohibitive for sparse harmful features.
Thm. 2	Alg. B offers substantially higher sample efficiency when the harmful witness lies in the ShapG neighborhood.
Prop. 2a/b	MinPPI errors affect only convergence speed: finite-$G_{\max}$ front stays internally non-dominated, optimal as $G_{\max} \to \infty$.

$f(\mathbf{m}) = (\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_\mathbf{m}),\ T_{\text{train}}(\mathcal{M}_\mathbf{m}))$ alone. Then (i) the selection outcome is independent of $\widehat{\text{MinPPI}}$; (ii) whenever the population size $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$, the entire first non-dominated front of $\mathcal{R}_g$ survives, so no f-dominated solution displaces an f-non-dominated one; and (iii) the returned set P_{out} is mutually non-dominated under f.

Proposition 2b (Asymptotic Optimality). If $\beta < \infty$—so bit-flip mutation equation 9 assigns every $\mathbf{m} \in \{0,1\}^n$ positive probability each generation—and selection is elitist, then as $G_{\max} \to \infty$ the returned front converges almost surely to the true Pareto front P^*_{true}.

Scope and implication. For finite $G_{\max}$ we guarantee only that P_{out} is internally non-dominated (Prop. 2a), *not* $P_{\text{out}} \subseteq P^*_{\text{true}}$; global optimality is the asymptotic statement (Prop. 2b). Within this scope MinPPI is a *safe* guidance signal: it shapes which candidates are proposed and how fast the front converges, while selection always defers to the true objective. Any importance method may thus be substituted without compromising the internal consistency of the returned front.

5.4 Summary

Table 3 consolidates the theoretical guarantees.

6 Experiments

We evaluate MOFS-MinPPI against seven baselines on five tabular classification datasets (Table 1): Spambase (56 feat., 4,601 samples), Madelon (100 feat., 2,600 samples), ISOLET (100 feat., 7,797 samples, 26 classes), Gisette (100 feat., 7,000 samples), and Credit-Card-Fraud (29 feat., 284,807 samples). Baselines are SHAP-Top-k, ShapG, MinShap, Banzhaf-FS, Powershap, MOFS-SHAP (SHAP-guided NSGA-II ablation), and NSGA2-uniform (unguided NSGA-II). MOFS-MinPPI uses $\alpha = 0.95$, $\beta = 1.0$, Algorithm B with $K_{\text{adv}} \in [8, 30]$ scaled per dataset, and an XGBoost fitness ($n_{\text{est.}} = 100$, depth 4) for fair HV and compression comparison. All methods use 5 seeds; means over seeds reported. Metrics: top-k AUC ($k \in \{1, \ldots, 50\}$) for all methods; Pareto-front HV and compression rate (CR$= 1-|F|/n$) at $1/3/5\%$ AUC-loss budgets additionally for NSGA-II methods.

6.1 Main Results

Table 4 reports AUC@10 (mean over 5 seeds). MOFS-MinPPI is competitive but not dominant—Pareto-front frequency ranking proxies per-feature importance, not single-feature predictive power. It leads on Credit-Card-Fraud (AUC@10=0.982).

Table 4: AUC@10 (mean over 5 seeds). Best per row in bold.

Dataset	MinPPI	SHAP-k	MOFS-S	NSGA2-u	Banzhaf	MinShap	ShapG	PShap
				AUC@10				
Spambase	0.904	**0.980**	**0.980**	0.973	0.963	0.970	0.965	**0.980**
Madelon	0.724	0.802	0.796	0.712	**0.821**	0.756	0.810	0.813
ISOLET	0.969	0.974	0.975	0.972	0.967	**0.975**	0.909	0.970
Gisette	0.950	**0.976**	**0.976**	0.954	0.957	0.958	0.947	0.966
Credit-Card	**0.982**	**0.984**	0.981	0.975	0.966	0.968	0.969	**0.983**

We report HV and CR@3%. MOFS-MinPPI achieves the highest HV on two datasets (ISOLET 36.44, Credit-Card 28.24) and the best CR@3% on Credit-Card-Fraud (91%). Friedman–Nemenyi on AUC@10 ($\chi^2 = 14.9$, $p = 0.037$) and AUC@20 ($\chi^2 = 17.6$, $p = 0.014$) confirms significant method differences; post-hoc places MOFS-MinPPI among the top tier.

6.2 Ablation

Min vs. mean operator: MinPPI delivers better AUC@10 on 4/5 datasets (e.g., spambase 0.948 vs. 0.893), with comparable HV. **Binary v_α vs. continuous**: binary yields structured harmful rates (1.8–27%, 0 decisive); continuous flags 57–100%—no discrimination. **Guidance β**: $\beta > 0$ never degrades HV; ISOLET improves from 0.078 to 0.114 at $\beta = 1$. **ShapG vs. SHAP**: ShapG detects more harmful on Madelon (32 vs. 23), SHAP leads on Credit-Card (9 vs. 3). **Alpha sensitivity** (Appendix A): harmful counts increase monotonically; decisive count zero (Corollary 1); $\alpha = 0.95$ balances signal (2–27%) vs. over-selectivity.

Author Contributions

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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A PROOFS

Proof of Property 1. For any $S \subseteq N_1$, the marginal $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$ takes values in $\{-1, 0, +1\}$ because $v_\alpha \in \{0, 1\}$. Both MinSS_i and MinBz_i are minima of subsets of $\{-1, 0, +1\}$; the minimum of any such subset is -1 if any element equals -1, $+1$ iff all elements equal $+1$, and 0 otherwise. Hence it remains in $\{-1, 0, +1\}$. □

Proof of Property 2. **(a)** ($\Rightarrow$) $\text{MinSS}_i = +1$ means every π yields marginal $+1$, forcing $v_\alpha(S_\pi^i) = 0$ and $v_\alpha(S_\pi^i \cup \{i\}) = 1$ for every π. As π runs over all permutations, S_π^i runs over all subsets of $N_1 \setminus \{i\}$ (each S appears as a prefix for $|S|!\,(n - |S| - 1)!$ permutations). Hence $\forall S \subseteq N_1 \setminus \{i\}$: $v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, every marginal is $1 - 0 = +1$, so $\min = +1$.

(b) $\text{MinSS}_i = -1$ iff there exists a permutation with -1 marginal, i.e., $v_\alpha(S_\pi^i) = 1$ and $v_\alpha(S_\pi^i \cup \{i\}) = 0$. Letting $S = S_\pi^i$ gives the witness. Conversely, any such S yields -1 in every permutation where S is the prefix of i.

(c) By Property 1, if MinSS_i is neither $+1$ nor -1, it must be 0. □

Proof of Property 3. $\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \Delta_i(S_\pi^i)$ and $\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \Delta_i(S)$. The collection $\{S_\pi^i : \pi \in \Pi(N_1)\}$ equals $2^{N_1 \setminus \{i\}}$, because each $S \subseteq N_1 \setminus \{i\}$ appears as a prefix for exactly $|S|!\,(n - |S| - 1)!$ permutations. Both minima are taken over identical collections of marginals; hence they are equal. □

Proof of Lemma 1. Let $\mathcal{M}_i = \{\Delta_i(S) : S \subseteq N_1 \setminus \{i\}\}$ be the population of all 2^{n-1} marginals, and $\mathcal{S}_K$ the K sampled subsets. The estimate $\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_{S \in \mathcal{S}_K} \Delta_i(S)$ satisfies $\min \mathcal{S}_K \geq \min \mathcal{M}_i$ deterministically because $\mathcal{S}_K \subseteq \mathcal{M}_i$. Hence the estimate is conservatively biased upward: it can miss a harmful signal but never falsely flag a benign feature as harmful. □

Proof of Proposition 1. Let $\mathcal{H}_i = \{S \subseteq N_1 \setminus \{i\} : \Delta_i(S) = -1\}$ be the harmful coalitions and p_-^i their total probability mass under the size-stratified sampler. Algorithm A fails iff none of the K i.i.d. draws lands in $\mathcal{H}_i$; independence gives $\Pr[\text{miss}] = (1 - p_-^i)^K$. Setting this $\leq \delta$ and solving yields the bound. □

Proof of Theorem 1. For each $i \in \mathcal{H}$, Proposition 1 gives $\Pr[\text{miss } i] = (1 - p_-^i)^K \leq (1 - p_{\min})^K$. By the union bound, $\Pr[\text{any miss}] \leq |\mathcal{H}| \cdot (1 - p_{\min})^K$. Setting this $\leq \delta$ and solving for K yields equation 10; the approximation uses $-\log(1 - p_{\min}) \approx p_{\min}$ for small $p_{\min}$. □

Proof of Theorem 2. By equation 11, $S_i^* \subseteq S_{\text{seed}}$. A single perturbation reproduces exactly S_i^* when every $j \in S_i^*$ is retained (probability $\prod_{j \in S_i^*} q_j$) and every $j \in S_{\text{seed}} \setminus S_i^*$ is dropped (probability $\prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j)$); these events are independent, so $\Pr[S = S_i^*] = \rho_i$. When $S = S_i^*$ the marginal is $v_\alpha(S_i^* \cup \{i\}) - v_\alpha(S_i^*) = -1$ and i is detected. Over K_{adv} independent perturbations, the probability of at least one hit is $1 - (1 - \rho_i)^{K_{\text{adv}}}$. If $S_{\text{seed}} = S_i^*$, the seed evaluated before any perturbation is already the witness, giving deterministic detection at $K_{\text{adv}} = 1$. □

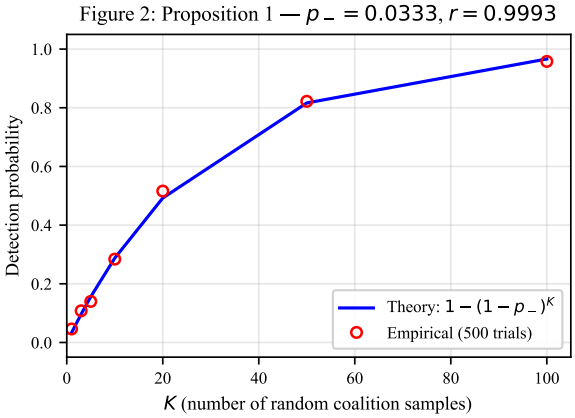


Figure 3: Empirical verification of Proposition 1 on an engineered voting game with $p_-^i = 0.0333$ (500 trials per K). Theory (solid line) and empirical rates (open circles) agree closely ($r = 0.9993$). For sparse harmful contexts ($p_- \ll 1$), Algorithm A requires $K \approx 90$ to reach 95% detection.

Table 5: Quantitative comparison of Algorithm A and Algorithm B (500 trials per setting).

(a) Detection of MinSS $= -1$

K	A detect	B detect	A $\bar{s}$	B $\bar{s}$
1	0.026	1.000	1.00	1.00
2	0.060	1.000	1.98	1.00
3	0.124	1.000	2.87	1.00
5	0.146	1.000	4.69	1.00
10	0.322	1.000	8.43	1.00
20	0.466	1.000	14.91	1.00
50	0.850	1.000	23.77	1.00

(b) Robustness to $\mathcal{N}(0, \sigma^2)$ on W (K=5)

σ	B detect	A detect
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148

Proof of Proposition 2a. $\widehat{\text{MinPPI}}$ enters only the generation operators (mutation equation 9, crossover), which determine *which* candidates populate $P_{\text{offspring}}$; once generated, a candidate is scored solely by f. Non-dominated sorting and crowding distance are functions of the f-values of $\mathcal{R}_g$ only, giving (i). The first front $\mathcal{F}_1(\mathcal{R}_g)$ ranks ahead of all dominated solutions; if $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$ it is admitted in full, so dominated solutions can occupy only the remaining slots, giving (ii). At termination $P_{\text{out}} \subseteq \mathcal{F}_1(\mathcal{R}_{G_{\max}})$, whose members are pairwise non-dominated, giving (iii). □

Proof of Proposition 2b. With $\beta < \infty$ the per-bit flip probability lies in $(0, 1)$, so from any state the offspring distribution places positive probability on every point of $\{0, 1\}^n$; the induced Markov chain over populations is therefore irreducible. Elitism guarantees that an f-non-dominated solution, once present, is never replaced by a solution it dominates (Prop. 2a(ii)). Irreducibility ensures every $\mathbf{m} \in P^*_{\text{true}}$ is generated in finite expected time and, being non-dominated, retained thereafter, while any dominated incumbent is displaced once a dominating point is sampled. Hence the front converges almost surely to P^*_{true}. The estimate $\widehat{\text{MinPPI}}$ alters the transition kernel but preserves its positivity, so it affects only the expected hitting times—the convergence speed—not the limit. □

B Empirical Verification of the Estimators

The toy-game verifications supporting Proposition 1 and the Algorithm A/B comparison are collected here.